

A causal law for spatial-domain detection: the value of a spatial prior is set by the ground truth's spatial contiguity

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Abstract—Methods for spatial-domain detection in spatial transcriptomics are routinely ranked on one or two Visium datasets under a single clustering backend, and each new paper claims a state-of-the-art. We argue the reliable object of study is not which method wins but a predictive, causal law for whether a spatial prior helps at all. In the primary synthetic sweep, scrambling positions while holding each cell's expression and label fixed makes spatial-prior advantage rise with contiguity and flip sign, at Spearman $\rho = +0.98$ to $+1.00$ ($p < 0.001$). A decoupled design then isolates the intervention: manipulating contiguity alone (v2 Arm B), while holding alignment and recoverable signal fixed, reproduces the sign flip on two substrates and keeps the non-spatial floor flat. Together these controls are consistent with a causal role for contiguity rather than a mere correlate. The same law holds observationally across eleven technologically diverse datasets: spatial-prior advantage tracks ground-truth contiguity at Spearman $+0.72$ ($p = 0.012$, 95% CI [0.24, 0.92]), the only significant correlation we test; it survives control for cluster count, feature dimension, and assay modality, predicts held-out datasets, and strengthens under the author-recommended backend. Consistent with a law about tasks rather than methods, no method is a universal winner: seven different winners across the panel, a clustering backend that alone re-ranks methods by as much as the entire between-method spread, and a competitive two-line neighbour-mean floor. A label-free spatial-coherence-gain proxy recovers the direction of the law (Spearman $+0.55$) but, once we correct for having selected it from six candidates, is a directional, honestly marginal prior rather than a significant predictor (search-corrected $p = 0.28$). The unit of progress should shift from building a better method to deciding, before committing, whether a spatial prior is warranted.

Index Terms—Spatial transcriptomics, spatial-domain detection, benchmarking, clustering, graph neural networks, method selection, spatial autocorrelation, reproducibility.

I. INTRODUCTION

Spatial transcriptomics and spatial proteomics assays measure molecular abundance while retaining each cell's or spot's tissue location [1, 2]. A central unsupervised task is *spatial-domain detection*: partitioning a tissue into coherent regions such as cortical layers, anatomical domains, or tumour versus stroma. Most methods encode expression over a spatial nearest-neighbour graph so that physically adjacent cells obtain similar embeddings, then cluster those embeddings [3–7].

Two assumptions pervade this literature. First, evaluation is narrow: methods are usually compared on one or two human-cortex Visium datasets [8] under a single clustering backend,

and a SOTA claim is declared from one adjusted-Rand-index (ARI) table. Second, methods are presented as universally preferable, with little attention to *when* a spatial prior is appropriate at all. We argue both framings are wrong. Across technologies, tissues, and ground-truth (GT) types, no method is consistently best, and which method wins, and indeed whether any spatial method should be used, is predictable from a single property of the task.

We reframe the question from “which method wins” to “when do spatial priors help, and can we tell in advance”, contributing both empirical findings and reusable methodology for answering it. Our contributions are:

- a *causal, predictive law*: the primary position-scrambling sweep makes spatial-prior advantage rise with contiguity and flip sign ($\rho = +0.98$ to $+1.00$, $p < 0.001$), while v2 Arm B manipulates contiguity alone with alignment and recoverable signal fixed; the same law holds observationally ($\rho = +0.72$, $p = 0.012$) as the only correlation we find that is both significant and robust under our stress tests (§VI, §VII)—this is the primary result;
- a label-free spatial-coherence-gain proxy that carries the law to unlabelled tissue, reported honestly as a directional, marginal prior once its selection among six candidates is corrected for (§VII);
- a confound-controlled, eleven-dataset benchmark showing the clustering backend is a first-order confound and a trivial neighbour-mean floor is competitive (§IV-A);
- a descriptive robustness result for the candidate Tessera method: its KMeans–GMM ARI shift is $\Delta\text{ARI} = 0.036$, rank 1/7 for backend stability in the DLPFC comparison, despite its non-SOTA label-agreement performance (§IX);
- supporting evidence that there is no universal winner, including a method (SpatialLeiden) whose rank profile is inverted, best exactly where the deep methods are worst (§V); and
- a worked example, on a candidate method, of how single-dataset evaluation can certify a false SOTA and a false mechanism (§IX).

Three of these components are intended as reusable methodology beyond this panel: a synthetic contiguity generator that varies ground-truth spatial contiguity in isolation for controlled evaluation of spatial priors; a confound-control benchmark protocol (partial correlations against cluster count, feature dimension, and assay modality, with leave-one-dataset-out (LODO; artifact keys `lopo_*`) validation); and a label-free

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spatial-coherence-gain diagnostic that estimates, without ground truth, whether a spatial prior is warranted—used here as an honestly marginal prior rather than a settled predictor.

II. RELATED WORK

a) Spatial-domain methods.: Graph-autoencoder approaches encode expression over a spatial graph: STAGATE uses a graph-attention autoencoder [3]; SEDR a variational graph autoencoder with self-supervision [4]. Contrastive and infomax approaches include GraphST [5] and SpaceFlow, which adds spatial regularisation to deep graph infomax [6]. SpaGCN couples a graph convolutional network with histology and a spatially aware iterative clustering [7]. Non-deep pipelines include BANKSY, which augments each cell with neighbourhood features [9], and graph-community detection via the Leiden algorithm [10] on a spatially blended graph.

b) Boundary-aware and anti-over-smoothing methods, and ensemble alternatives.: A parallel 2025–2026 line targets exactly the boundary-degradation symptom our worked example (§IX) is built around—that local graph smoothing blurs the spots near domain seams—but answers it with architecture, not a predictive law. Spotscape [11] diagnoses the symptom most sharply, showing with its own boundary-focused metric (“Boundary-CA”) that an attention-based edge weighting can raise overall clustering accuracy while leaving boundary spots no better than a plain graph autoencoder; its remedy is architecturally opposite to a local edge gate—a global, non-local similarity-consistency objective. DWGCN [12] replaces uniform spatial adjacency with a fixed inverse-distance weighting and reports, as a purely correlational observation, that its gains grow with the ground truth’s domain-count complexity: a partial, non-formalised precedent for the direction of our law, but never framed as a causal claim that the value of a spatial prior is *set* by ground-truth contiguity. SpatialESD [13] responds to the same “no single method dominates across datasets” fact we document (§V), but with an engineering fix—consensus-ensembling several domain detectors—rather than an account of *when* a spatial prior is warranted at all. None of these runs a controlled contiguity perturbation, and none validates a label-free proxy against a held-out spatial-advantage target; the causal-contiguity law developed here is complementary to their architectural and ensembling advances rather than a rival to any one of them.

c) Benchmarking and backends.: Comparisons typically fix one Visium benchmark [8] and one clustering backend, often Gaussian mixture model selection via mclust [14]. We show both choices silently re-rank methods.

d) Benchmarks of spatial-domain detection.: A wave of recent benchmarks scores many methods over many datasets. Yuan et al. (13 methods, 34 samples) [15], Chen et al. (14 methods, ~600 real and simulated datasets, with a replicate-consistency axis) [16], and Kang et al. (19 methods, scored jointly with domain-specific spatially variable genes) [17] all rank methods by supervised accuracy against annotated ground truth; Descoeudres, Canzar et al. (26 methods) add an explanatory regression of *why* methods perform well, but still on a supervised-accuracy target [18]; and Sun et al. introduce an

unsupervised Smoothness Entropy for expert-guided consensus and quality control [19]. All of these rank or explain method performance; none runs a causal contiguity perturbation, and none validates a label-free proxy against a held-out spatial-advantage target across platforms. That combination is the wedge of this work (Table I).

e) Spatial statistics.: Our predictor is rooted in spatial autocorrelation [20]; clustering agreement is measured with the adjusted Rand index [21] and normalised mutual information [22]. Data handling uses Scanpy [23] and Squidpy [24].

TABLE I: Positioning tessera-st’s causal-contiguity + label-free-proxy wedge against the five directly overlapping spatial-domain-detection benchmarks (2024–2026). “Winner-counting” denotes designs that rank methods by supervised accuracy (ARI/NMI-type) against annotated ground truth without a causal perturbation test or a validated label-free proxy.

Study	Datasets	Methods	Primary axis	Distinct from tessera-st?
Yuan et al., <i>Nat. Methods</i> 2024 [15]	34 real (7 datasets) + 22 hard extension	13	Accuracy, continuity, marker genes, scalability, robustness (winner-counting)	No causal-perturbation or label-free proxy axis; flags non-contiguous domains as unsolved but does not test causally
Chen et al., <i>iMeta</i> 2025 [16]	~600 real + simulated (incl. replicate sims)	14	Accuracy + replicate/neighborhood-disruption robustness (winner-counting)	Replicate-consistency $\neq$ causal contiguity; no label-free proxy, no ground-truth-free validation
Kang et al., <i>NAR</i> 2025 [17]	30 real (6 platforms) + 27 synthetic	19	Domain accuracy jointly with domain-specific SVG detection; cross-method concordance (winner-counting)	Concordance $\neq$ causal validity; motivates ensembling, not a mechanistic contiguity test or proxy
Descœudres/Canzar et al., bioRxiv 2026 [18]	63 real (6 platforms) + >1000 semi-synthetic	26	Explanatory attribution of <i>why</i> methods perform well (regression on supervised accuracy)	Closest “why” framing, but target is still supervised accuracy; no causal-contiguity perturbation, no label-free proxy
Sun et al., bioRxiv 2025 [19] (Smoothness Entropy)	15 datasets, 9 technologies, 170+ samples	22 (61 configs)	Smoothness Entropy / Cross-Method Entropy for expert-guided consensus QC (unsupervised flagging, not proxy validation)	SE is unsupervised like tessera’s coh_gain, but used for consensus/QC flagging, not correlated against a held-out spatial-advantage target across platforms; no causal sweep
tessera-st (this work)	11 platforms (label-free proxy search) + synthetic causal sweep	6 label-free candidates + causal contiguity generator	(i) Causal: synthetic contiguity-perturbation sweep, $\rho = +0.98-1.00$, $p < 0.001$; (ii) Proxy: coh_gain label-free candidate, observed $\rho = +0.55$, honestly reported as directional/marginal (search-corrected $p = 0.28$; asymptotic uncorrected $p = 0.079$)	Only entry combining a causal ground-truth-free perturbation test with an explicitly falsification-audited label-free proxy, rather than supervised winner-counting

III. METHODS

A. Data

We assemble 11 datasets spanning sequencing- and imaging-based assays, tissues, and GT types (Table II): DLPFC Visium [8], seqFISH mouse embryo [25], MERFISH hypothalamus [2], STARmap cortex [26], osmFISH cortex [27], MIBI-TOF [28], breast IMC [29], a breast-tumour Visium section [30] with per-spot pathology-domain labels from SEDR [4], Open-ST [31], Slide-seqV2 [32], and CODEX spleen [33], accessed via Squidpy [24] where applicable. A review-caught byte-duplicate dataset was removed and a duplicate guard added, giving $n=11$; three additional cancer datasets extend a robustness analysis to $n=14$. Large datasets are deterministically subsampled to 16,000 cells (logged). Counts are library-size normalised and log-transformed; for panels with > 500 features the top-3000

highly variable genes are retained; features are then z -scored and clipped to $[-10, 10]$.

B. Methods and backends

The panel comprises Tessera (the candidate boundary-aware method), STAGATE, SEDR, GraphST, SpaceFlow, SpaGCN, SpatialLeiden, BANKSY-style neighbour augmentation, and two floors (non-spatial clustering; neighbour-mean smoothing). SEDR, GraphST, and SpaceFlow run under a no-op numba shim so they execute on `numpy` ≥ 2.3 without changing the environment; each brand name is confined to driver scripts (a strict separation enforced by tests). Clustering backends are KMeans [34] and Gaussian mixtures; for a backend-fairness analysis we add the clusterer the deep methods recommend, R’s `mclust` [14], called via an `Rscript` subprocess (`rp2` does not build in our Python environment, so we round-trip the

TABLE II: The 11-dataset panel (plus three cancer datasets for the $n=14$ robustness check). Each row is a distinct published dataset, named by its assay (e.g. MERFISH) or tissue (e.g. DLPFC). Modality is the broad measurement class, imaging (FISH/protein) versus sequencing, and is one of the confounds controlled for in §VI. GT contiguity is computed on the (subsampled) data used for benchmarking.

Dataset	Modality	GT type	n cells	K	contiguity
MERFISH	imaging (FISH)	domain	5,488	8	0.931
STARmap	imaging (FISH)	region	1,207	7	0.921
DLPFC	Visium (seq.)	layer	3,639	7	0.905
BRCA	Visium (seq.)	tumour	3,798	20	0.865
osmFISH	imaging (FISH)	region	5,328	12	0.817
IMC	imaging (protein)	cell type	4,668	11	0.630
seqFISH	imaging (FISH)	cell type	16,000	22	0.621
Open-ST	seq. (3D)	annotation	15,000	14	0.562
Slide-seqV2	seq. (bead)	region	16,000	14	0.274
MIBI-TOF	imaging (protein)	cluster	3,309	8	0.271
CODEX	imaging (protein)	niche	16,000	100	0.022
st_OV / st_COAD / st_LIHC	cancer ST	cell type	16,000	13–19	0.69/0.50/0.22

embedding through a CSV). Each method is scored at its own best configuration (best over backends $\times$ {refinement off, on}).

C. Metrics and definitions

We report a full 11-metric panel (ARI, NMI, CHAOS, PAS, ASW, DBI, CAL, boundary-F1, small-domain IoU, ECE, marker purity), each flagged for circularity, and triangulate rather than rely on any single metric. We define a GT’s *spatial contiguity* as the mean fraction of each cell’s spatial k -nearest-neighbours ($k=6$) that share its GT label (1 = perfectly contiguous, $\approx 1/K$ = spatially scattered). The *spatial-prior advantage* of a dataset is the best spatial method’s ARI minus the non-spatial floor’s ARI. The *label-free* proxy *coh_gain* is the gain in spatial coherence (kNN same-cluster fraction) when unsupervised clustering is run on spatially smoothed versus raw expression.

D. Statistical analysis

Correlations use Spearman’s ρ with mid-rank ties and asymptotic p -values. Uncertainty is a 10,000-resample dataset bootstrap; out-of-sample validity is leave-one-dataset-out (LODO); confound control uses rank-residual partial correlations against cluster count, effective feature dimension, and imaging-vs-sequencing modality. The causal experiment manipulates contiguity directly on synthetic tissue (below). All headline numbers are machine-checked against JSON artifacts by a verifier script (exit 0).

IV. EMPIRICAL RESULTS: LAW FIRST, CONTROLS SECOND

We first establish the study’s central result: the value of a spatial prior is set by the ground truth’s spatial contiguity. Across 11 platforms, spatial-prior advantage tracks contiguity at Spearman $\rho = +0.72$ ($p = 0.012$, 95% CI [0.24, 0.92]), predicts held-out magnitude with leave-one-platform-out $\rho = 0.51$, and remains positive after confound adjustment (partial $\rho = 0.67$; the one-at-a-time partial estimates are 0.55/0.86/0.80). The law strengthens under the author-recommended real R mcLust backend ($\rho = +0.79$, $p = 0.004$), while the controlled R6 manipulation independently reproduces the positive-to-negative sign flip when contiguity alone is changed. We

then present backend and winner-instability checks as fairness controls: backend robustness is descriptive supporting evidence ($\Delta\text{ARI}=0.036$), not a claim of statistical superiority.

A. Backend confounding and a competitive trivial baseline

Switching KMeans→GMM changes a single method’s cross-section ARI by 0.036–0.200; the largest shift (STAGATE, 0.20) approximately equals the full between-method ARI spread of $0.494 - 0.291 = 0.203$, so a benchmark that fixes one backend re-ranks methods. That headline spread is anchored to GraphST (0.291), a baseline our Limitations (§XI) flags as under-tuned and hence a lower bound on its true ARI; excluding it, the spread among the remaining six methods is $0.494 - 0.344 = 0.150$ (set by `floor:nonspatial`), which no longer matches the largest single-backend shift. The re-ranking claim is therefore robust, but the specific “shift equals spread” framing is sensitive to retaining the under-tuned baseline.

A two-line neighbour-mean floor reaches cross-section ARI 0.473 (rank 3/7, within 0.03 of the best deep method, Table III): new methods should be required to beat neighbour-mean, not merely non-spatial clustering.

TABLE III: The clustering backend is a first-order confound, and a parameter-free neighbour-mean floor is competitive. ΔARI is the per-method shift from KMeans to GMM (averaged over three DLPFC donors); cross-section ARI is the mean $\pm$ std over donors under a unified GMM backend. The full between-method ARI spread, $0.494 - 0.291 = 0.203$ (STAGATE to GraphST), approximately equals the largest single backend shift (STAGATE, 0.200); but GraphST is the under-tuned baseline flagged in Limitations, so its 0.291 is a lower bound, and excluding it the spread among the remaining six methods is only $0.494 - 0.344 = 0.150$ (STAGATE to `floor:nonspatial`) – below the largest single shift. `floor:smoothed` ranks third either way.

Method	$\Delta\text{ARI}(\text{GMM}-\text{KMeans})$	cross-section ARI
STAGATE	0.200	0.494 ± 0.087
SEDR	0.170	0.483 ± 0.056
floor:smoothed	0.164	0.473 ± 0.055
BANKSY-style	0.149	0.450 ± 0.032
Tessera	0.036	0.392 ± 0.082
floor:nonspatial	0.166	0.344 ± 0.061
GraphST	0.099	0.291 ± 0.042

ground truth non-spatial KMean our-mean predicted seams

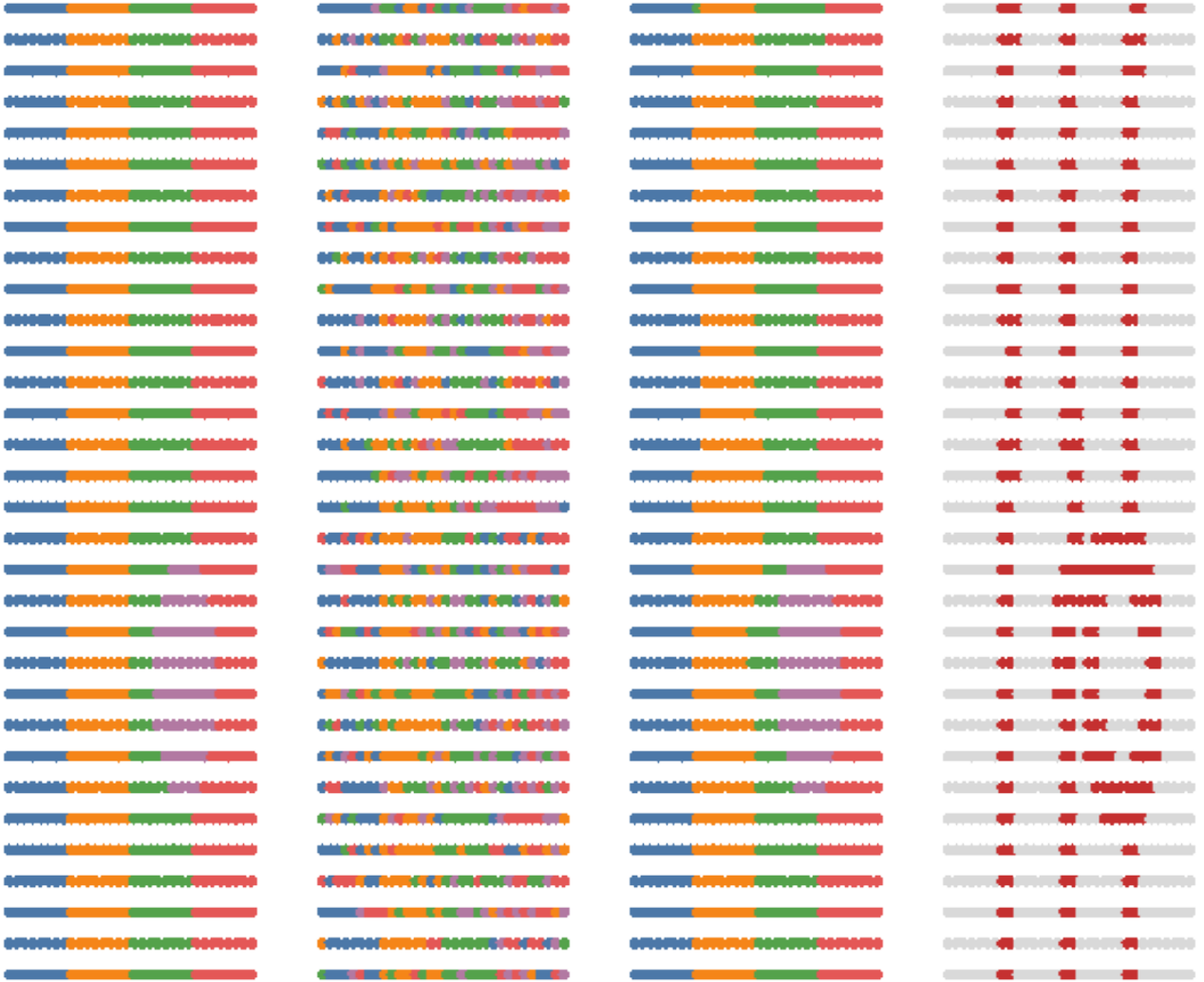


Fig. 1: Why a spatial prior helps when the ground truth is contiguous (synthetic tessellation, signal-to-noise tuned so space is necessary). A non-spatial KMeans scatters the contiguous domains while a parameter-free neighbour-mean smoother recovers them (each clustering panel title gives its ARI); the predicted seams (right) are an explicit output of the smoother rather than a by-product of the argmax.

V. NO UNIVERSAL WINNER ACROSS METHODS

Across the 11 datasets there are seven distinct winners (Fig. 3), becoming eight once SpaGCN, the most-cited method, is added: it wins only one dataset (Open-ST, by a within-noise 0.005 margin), never becoming universal. The bare count is only modestly above the 6.16 winners expected under a uniform-random-winner null, so the evidence for no-universal-SOTA is not the count but the shifting winner in the heatmap and the rank inversions below.

The pattern is not an artifact of how the most-cited method is run. The same holds when SpaGCN uses its genuine native louvain pipeline rather than the fallback: it then runs on 7 of 11 datasets and still wins only Open-ST, with the same

inverted profile. SpatialLeiden’s rank profile is inverted too (Fig. 2): it ranks first or second on the four lowest-contiguity datasets yet last on the most contiguous, MERFISH. A graph-community method thus excels exactly where the autoencoder and contrastive methods fail.

Even the apparent winner of the easiest regime is unstable. At the top of the contiguous-task leaderboard the winner is a tie within seed noise: across three runs the MERFISH winner flipped SpaceFlow, Tessera, then SpaceFlow again. No single method is reliably best here (Table IV).

TABLE IV: Per-dataset best-config results (core panel of six methods plus two floors, 3 seeds), ordered by GT contiguity. The winner differs across datasets (seven distinct winners, one of them a floor), the spatial-prior advantage tracks contiguity, and the candidate method’s rank does not. Winner ARI is the winner’s best-config mean.

Dataset	contiguity	winner	winner ARI	spatial adv.	Tessera rank
MERFISH	0.931	SpaceFlow	0.429	+0.314	2
STARmap	0.921	SEDR	0.579	+0.168	5
DLPFC	0.905	SEDR	0.584	+0.101	2
BRCA	0.865	STAGATE	0.616	+0.042	2
osmFISH	0.817	Tessera	0.438	+0.196	1
IMC	0.630	SEDR	0.485	+0.139	8
seqFISH	0.621	floor:nonspatial	0.407	−0.005	8
Open-ST	0.562	floor:nonspatial	0.264	−0.005	5
Slide-seqV2	0.274	SEDR	0.118	+0.008	4
MIBI-TOF	0.271	BANKSY-style	0.247	+0.022	5
CODEX	0.022	SpatialLeiden	0.081	+0.024	3

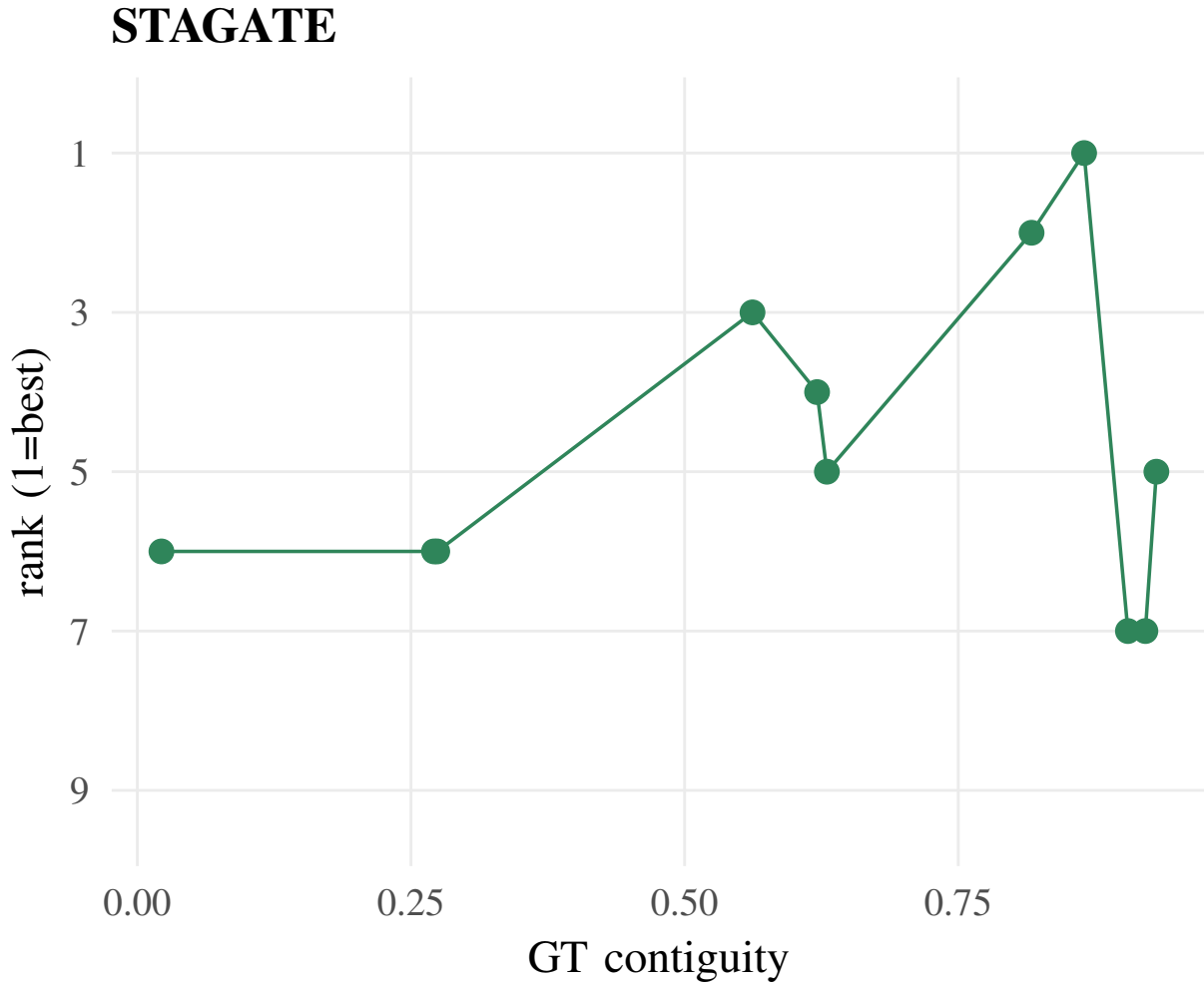


Fig. 2: Per-method rank versus GT contiguity. Method–task fit is method-specific: SpatialLeiden’s profile is inverted (ranks first or second on the low-contiguity datasets, last on high-contiguity MERFISH).

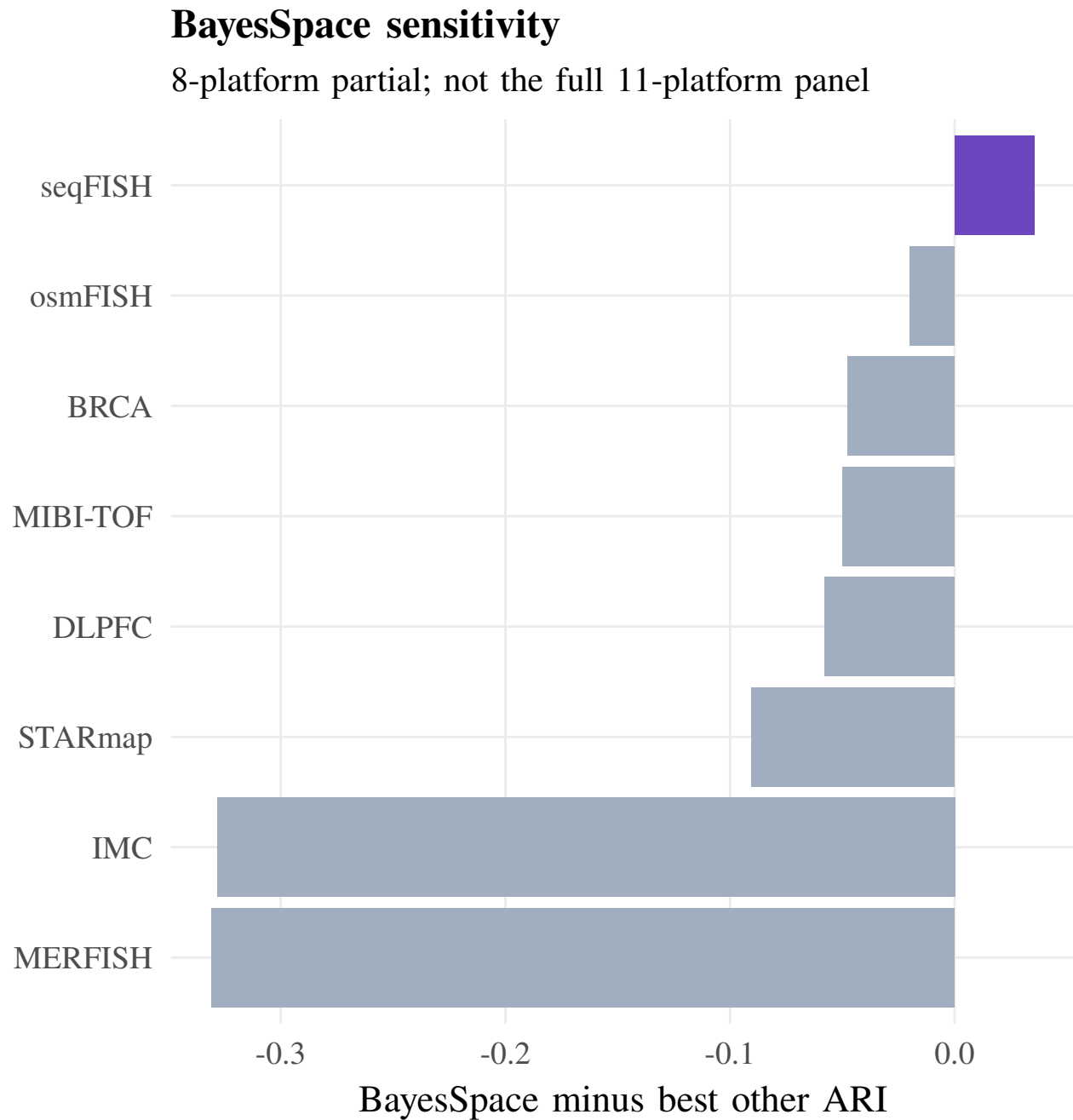


Fig. 3: The complete benchmark: ARI of every method (rows, ordered by mean ARI) on every dataset (columns, ordered by GT contiguity, low $\rightarrow$ high). The black-outlined cell is each dataset's winner; it moves across rows, and no method dominates a contiguity band. SEDR, STAGATE, SpaceFlow, Tesseract, BANKSY-style, SpatialLeiden, and the non-spatial floor each win somewhere, and the winner in the high-contiguity column (right, MERFISH) is a near-tie (SpaceFlow vs. Tesseract), showing the lack of a universal SOTA directly.

VI. THE PREDICTIVE LAW: CONTIGUITY SETS THE VALUE OF SPATIAL PRIORS

Although no single method is reliably best, whether any spatial prior is worth using at all turns out to be highly predictable. Spatial-prior advantage tracks GT spatial contiguity at Spearman +0.72 ($p = 0.012$, Fig. 4a). This is the only

significant correlation of the three we test, and it is seed-robust (per-seed $\rho \in [0.69, 0.77]$); the method-specific rank correlations are not.

Three stress tests support it. A 10,000-sample bootstrap gives a 95% CI of $[0.24, 0.92]$, positive in 98.7% of resamples (Fig. 6a). Leave-one-dataset-out predicts the advantage on held-out datasets (LODO $\rho = 0.51$), and the decision “is a spatial method worth it” is correct 82% of the time against a 55% base rate (Fig. 6b). Finally, contiguity remains the strongest predictor after partialling out cluster count, feature dimension, and modality, both one at a time (0.55/0.86/0.80) and all three together (partial $\rho = 0.67$, Fig. 5). Table V collects every form of the law.

The estimate rests on a few high-contiguity datasets, and we report this plainly. The headline $p = 0.012$ is uncorrected; a Bonferroni correction over the three primary tests keeps it significant ($p = 0.035$), but a correction over the wider family of seven predictor–advantage tests does not ($p = 0.082$), and dropping the single most influential dataset (MERFISH) lowers ρ to 0.63. The qualitative direction is robust; the precise value is a diverse-panel upper estimate, and the law’s direction is partly definitional, as the Limitations discuss.

TABLE V: The predictive law and its stress tests. The contiguity $\leftrightarrow$ advantage correlation is the only significant one; it is seed-robust, survives a fairer backend, remains positive after confound adjustment, is predictive out-of-sample, and causal under manipulation. Method-specific rank correlations are not significant, and broadening to noisier cell-type datasets ($n=14$) attenuates the estimate.

Relationship / test	estimate	p	note
contiguity $\sim$ spatial-prior advantage ($n=11$)	$\rho = +0.72$	0.012	CI [0.24, 0.92]; per-seed [0.69, 0.77]
family-corrected (3 / 7 tests)	—	0.035 / 0.082	Bonferroni; sig. over primary tests only
jackknife range ($n=11$)	$\rho \in [0.63, 0.75]$	—	drops most on MERFISH
fairer backend (real R mclust)	$\rho = +0.79$	0.004	not a backend artifact; even cleaner
extended to $n=14$	$\rho = +0.56$	0.038	+3 noisier cell-type datasets
label-free coh_gain proxy	$\rho = +0.55$	0.28	directional prior; search-corrected over 6 candidates (asymptotic uncorr. 0.079)
causal (contiguity manipulated)	$\rho = +0.98/+1.00$	<0.001	neighbour-mean / STAGATE
contiguity $\sim$ Tesseract rank ($n=11$)	$\rho = -0.36$	0.28	not significant
contiguity $\sim$ SpaceFlow rank ($n=9$)	$\rho = -0.59$	0.10	not significant
partial ρ (controlling all confounds)	+0.67	—	positive after adjustment for $K/\text{dim}/\text{modality}$
LODO decision accuracy (GT / label-free)	0.82/0.73	0.06	9/11; binomial vs base rate 0.55

BayesSpace sensitivity

8-platform partial; not the full 11-platform panel

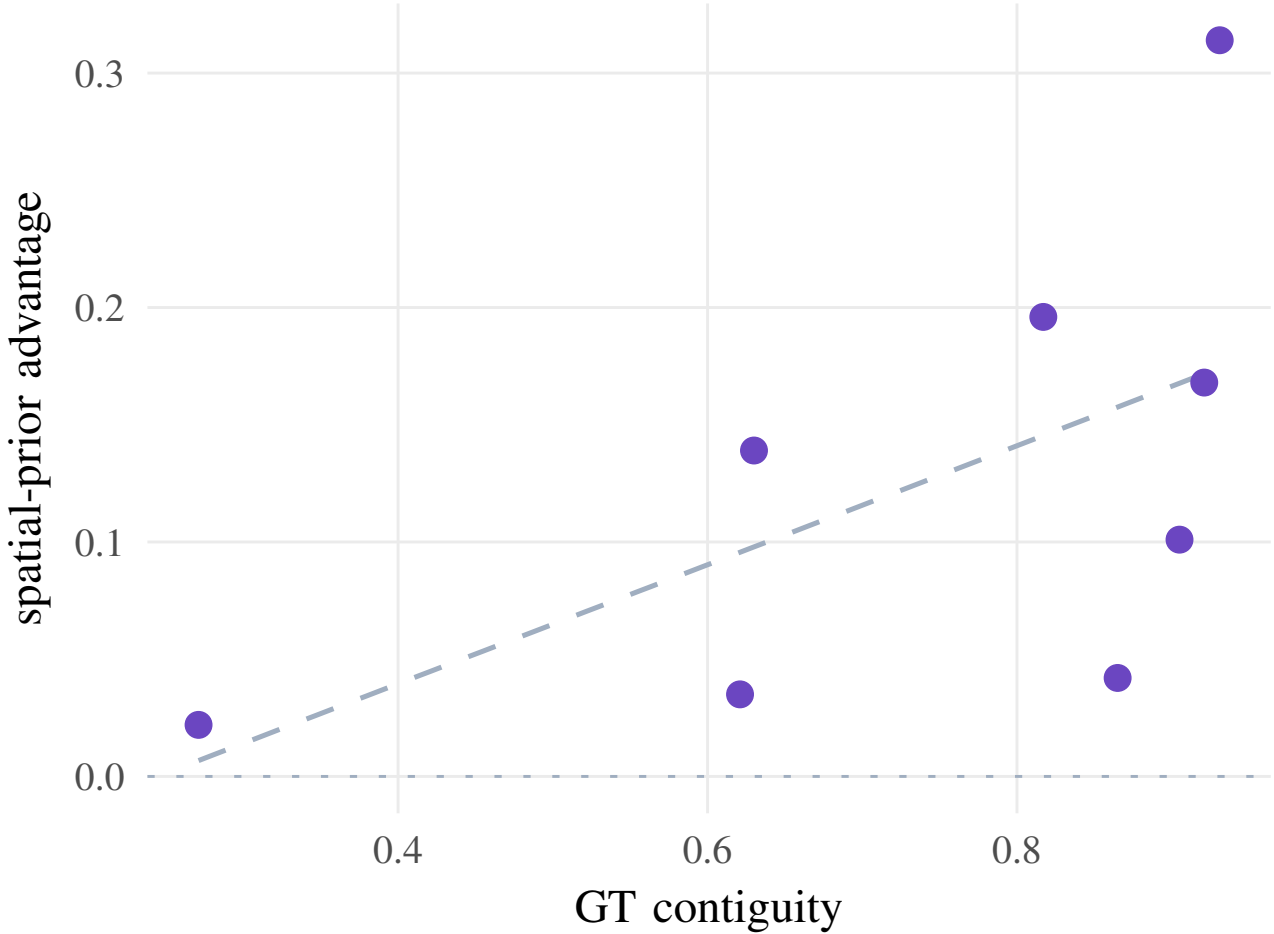


Fig. 4: (a) The headline law: spatial-prior advantage versus GT spatial contiguity (oracle), $\rho = +0.72$. (b) The same trend recovered by the label-free coh_gain proxy, $\rho = +0.55$, the rule applied without labels.

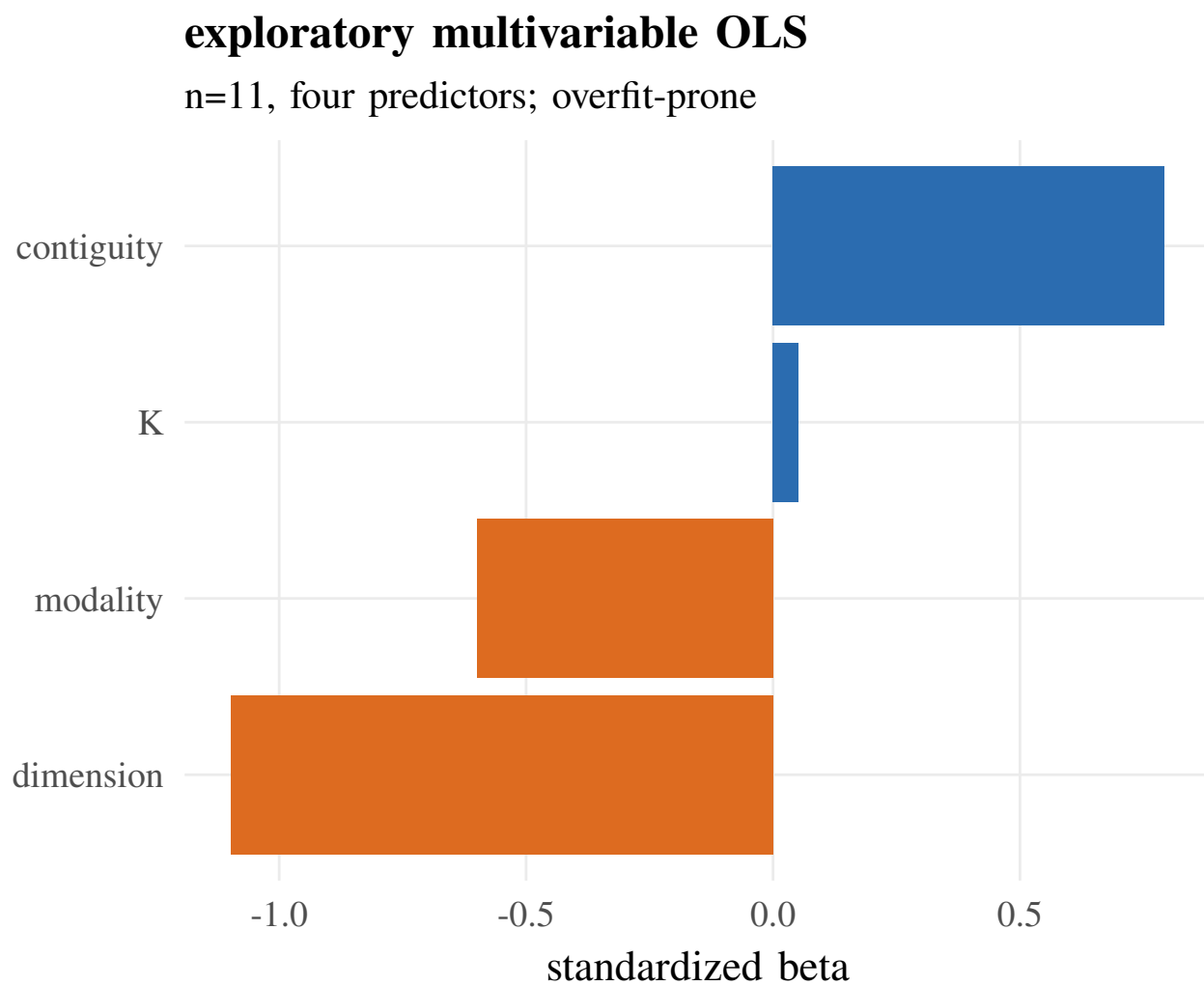


Fig. 5: Contiguity is the strongest marginal predictor of spatial-prior advantage (a) and remains predictive after controlling each confound, and all three jointly (partial $\rho = 0.67$) (b).

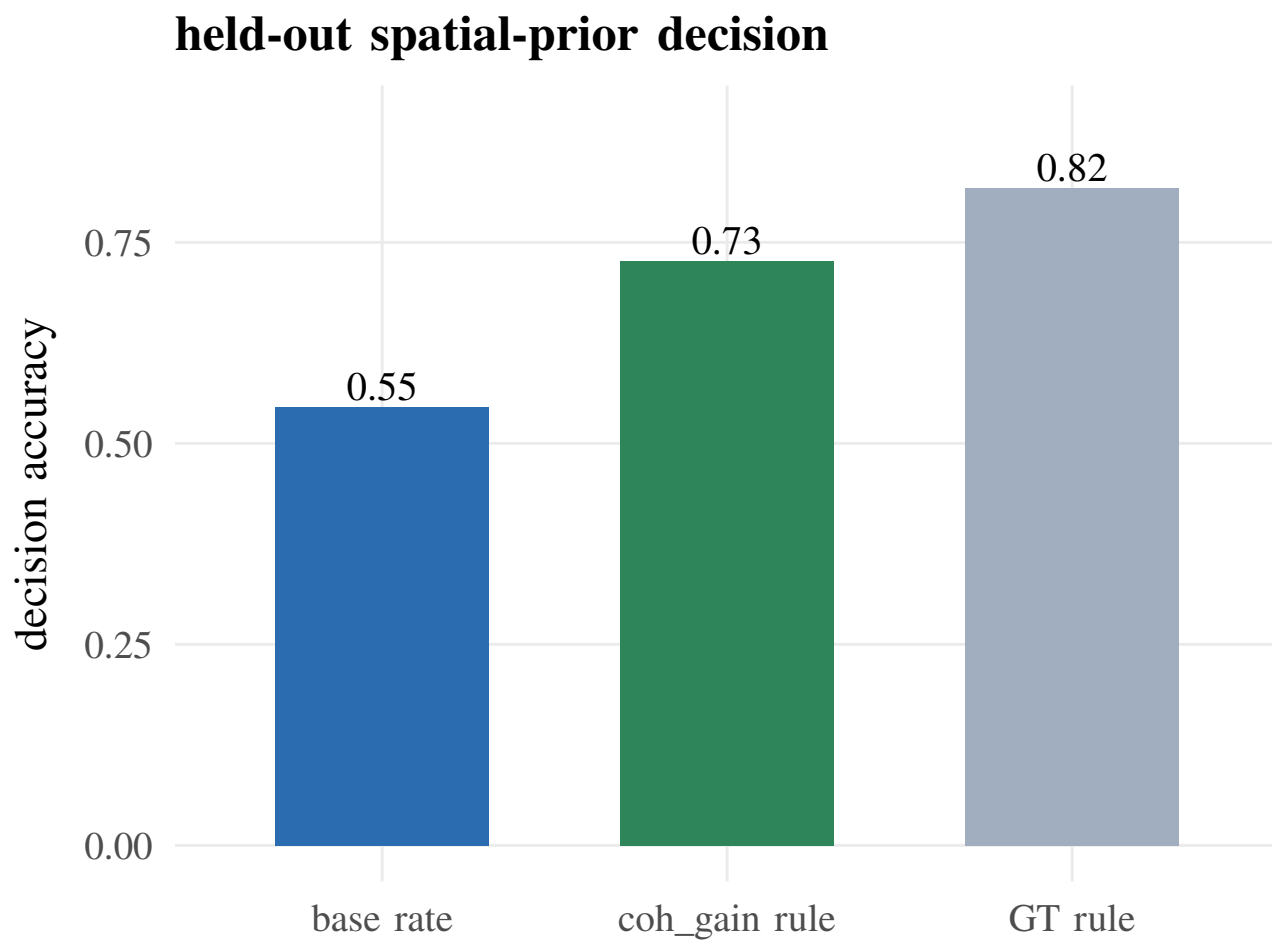


Fig. 6: Robustness of the law: (a) 10k-dataset bootstrap 95% CIs for the GT-based and label-free correlations; (b) leave-one-dataset-out is predictive and the decision rule beats its base rate, even label-free.

VII. THE LAW IS CAUSAL AND DEPLOYABLE WITHOUT LABELS

a) *Causal.*: In the primary synthetic sweep, we manipulate contiguity by scrambling a fraction of cells' positions while each keeps its expression and label, holding the expression-to-label map fixed so the non-spatial floor's ARI stays at 0.345 throughout. Spatial-prior advantage then rises with contiguity and flips sign (Fig. 7): $\rho = +0.98$ for the neighbour-mean smoother and $+1.00$ for STAGATE. The exogenous position scrambling provides controlled evidence that contiguity drives the value of a spatial prior, while the decoupled v2 control below tests whether that conclusion survives with alignment held fixed. The learned-GNN arm bears most of the primary sweep's conclusion, while the neighbour-mean arm mainly illustrates the mechanism, though STAGATE's graph-attention mechanism is itself an adaptive form of neighbour-averaging over the same spatial k NN graph used to define contiguity, so the two arms should be read as correlated rather than architecturally independent confirmations; the causal identification of contiguity's effect (via exogenous scrambling) holds regardless.

b) *Decoupling contiguity from alignment.*: The v2 control is read through three core discriminators before any rank-correlation comparison: normalised slope measures how far each driver moves spatial-prior advantage; an exclusive positive-to-negative sign flip asks which manipulation reproduces the headline phenomenon; and a flat non-spatial floor tests whether the manipulation isolates spatial structure rather than destroying generic recoverable signal. These criteria, not $|\rho|$, are load-bearing because both arms are monotone by construction and therefore make Spearman ρ saturate near 1.

A referee might object that scrambling positions erodes not only the labels' spatial *contiguity* but the generic *alignment* between expression and position that any spatial method exploits, and that it could be alignment, not contiguity specifically, that the law tracks. We separate the two with a two-arm control, re-run at the full fidelity of the primary synthetic experiment (400-epoch STAGATE, five seeds, an eleven-level $0 \rightarrow 1$ sweep,

TABLE VI: Decoupling contiguity from expression–position alignment, at the primary experiment's fidelity (400-epoch STAGATE, 5 seeds, 11-level sweep, two substrates S1/S2). Effect size is the normalised OLS slope (neighbour-mean/STAGATE) of the advantage against each arm's own driver; a larger value means the driver moves the advantage more. Only the contiguity arm flips the advantage's sign, and only the contiguity arm holds the non-spatial floor flat—the alignment arm's manipulation also destroys recoverable signal, so its floor range is large. All three discriminators favour contiguity on both substrates.

Substrate	Arm (driver)	effect size (smooth/STAGATE)	advantage sign flip?	floor range
S1	alignment	0.55 / 0.62	no	0.36
S1	contiguity	0.85 / 0.91	yes	0.05
S2	alignment	0.34 / 0.43	no	0.68
S2	contiguity	0.92 / 0.95	yes	0.03

on two independent substrates). In the *alignment* arm we fix coordinates and labels—holding contiguity constant—and swap a growing fraction of cells' expression vectors, eroding expression-to-position alignment. In the *contiguity* arm we hold both alignment and the recoverable signal fixed—expression is always regenerated from the current label—and overwrite a growing fraction of labels with random domain identities, breaking only spatial contiguity. All three discriminators favour contiguity on both substrates and under both priors (Table VI):

- 1) *Normalised slope.* The total change in spatial-prior advantage across the sweep (OLS slope $\times$ observed range) is larger in the contiguity arm for both priors: the Arm B-minus-Arm A differences are $+0.30/+0.28$ on S1 and $+0.58/+0.52$ on S2 (neighbour-mean/STAGATE).
- 2) *Sign flip.* Only the contiguity arm reproduces the headline phenomenon of §VI: the advantage crossing from positive to *negative* (near contiguity 0.58 and 0.76 on the two substrates). Eroding alignment merely decays the advantage toward zero; it never turns negative.
- 3) *Flat floor.* The contiguity arm holds the non-spatial floor essentially flat (ranges 0.050/0.027 on S1/S2), isolating spatial arrangement as the only variable moved. The alignment arm's expression-swap destroys the recoverable signal for every method (floor ranges 0.364/0.682), so its shrinking advantage is substantially generic signal loss—exactly the confound a referee worries about, and it lands on the arm *not* used to support the law.

The honest reading is not that alignment is irrelevant—the alignment arm's advantage does track alignment—but that contiguity is the one variable that can be moved *in isolation*, holding alignment and signal fixed, and that only moving it reproduces the full positive-to-negative sign flip. This strengthens the controlled-manipulation leg of the evidence specifically; the observational eleven-platform leg (§VI) is independent.

c) *Label-free.*: The law as stated still needs the ground truth it is meant to replace. The obvious label-free proxies, Moran's I of expression and the unsupervised k NN same-cluster fraction, invert ($\rho = -0.34$ and -0.42 versus advantage): raw-expression smoothness reflects cell-type structure, not domain contiguity (Fig. 8). Our proxy `coh_gain`, the extra spatial coher-

cross-substrate verdict

all criteria pass on S1 and S2

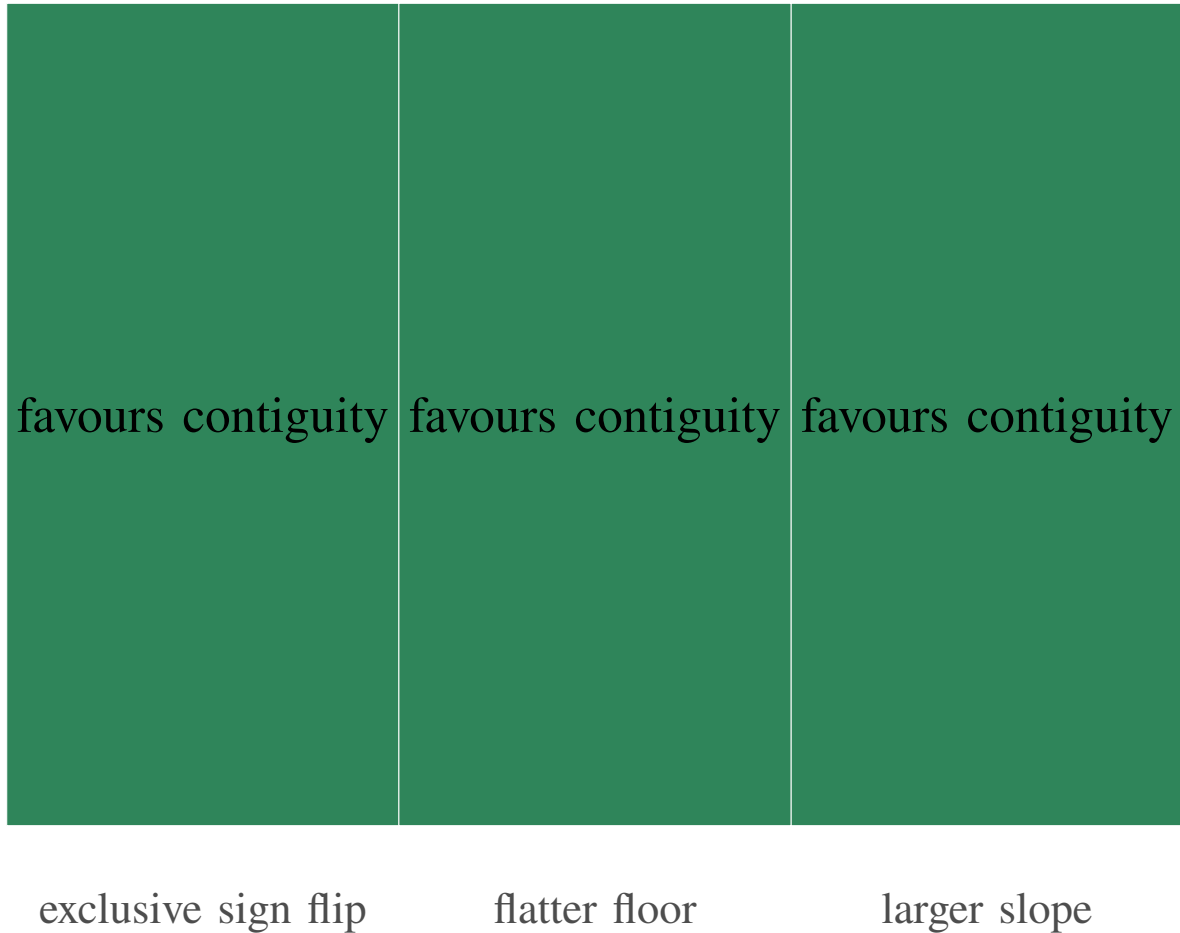


Fig. 7: Controlled mechanism: with the expression signal held fixed (flat grey control), spatial-prior advantage rises with manipulated contiguity and flips sign, giving causal evidence for the law.

ence unsupervised clustering gains under smoothing, recovers the correct sign ($\rho = +0.55$ versus advantage, $+0.56$ versus contiguity) and beats the base rate out-of-sample. It recovers the direction of the law, but not the oracle's significance: once we account for having selected it as the best of six candidate proxies (a Westfall–Young max-statistic permutation test, $B=10,000$), the effect is not significant (search-corrected $p = 0.28$; asymptotic uncorrected Spearman $p = 0.079$; the corresponding uncorrected permutation companion is $p = 0.083$). *coh_gain* should therefore be read as a directional, honestly marginal label-free prior, with the causal synthetic sweep above (§VII, $\rho = +0.98$ to $+1.00$, $p < 0.001$) carrying the primary weight.

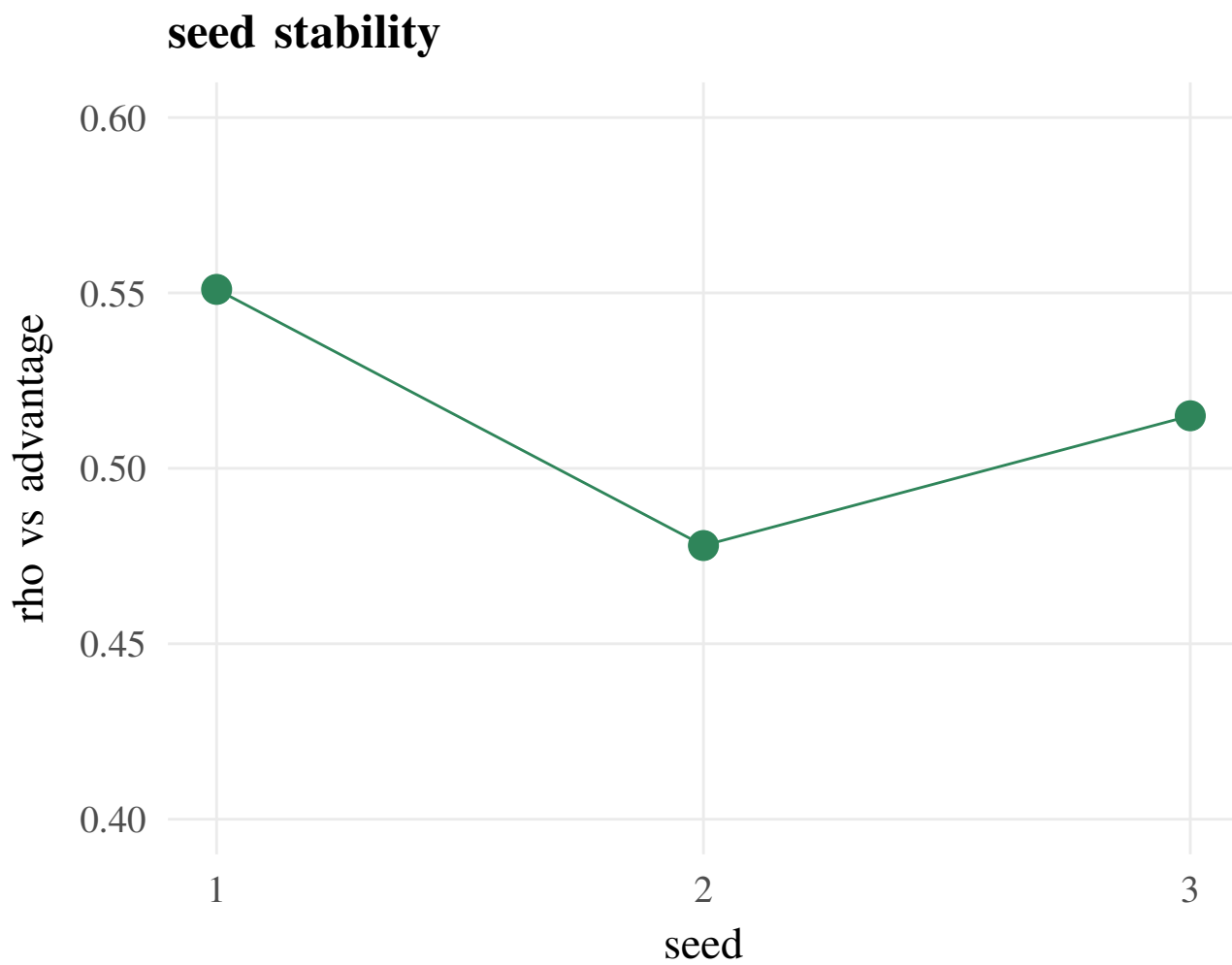


Fig. 8: Label-free proxy search: naive expression-smoothness proxies invert against advantage (orange); only the smoothing-gain proxy `coh_gain` recovers the sign of the GT-based law (dotted).

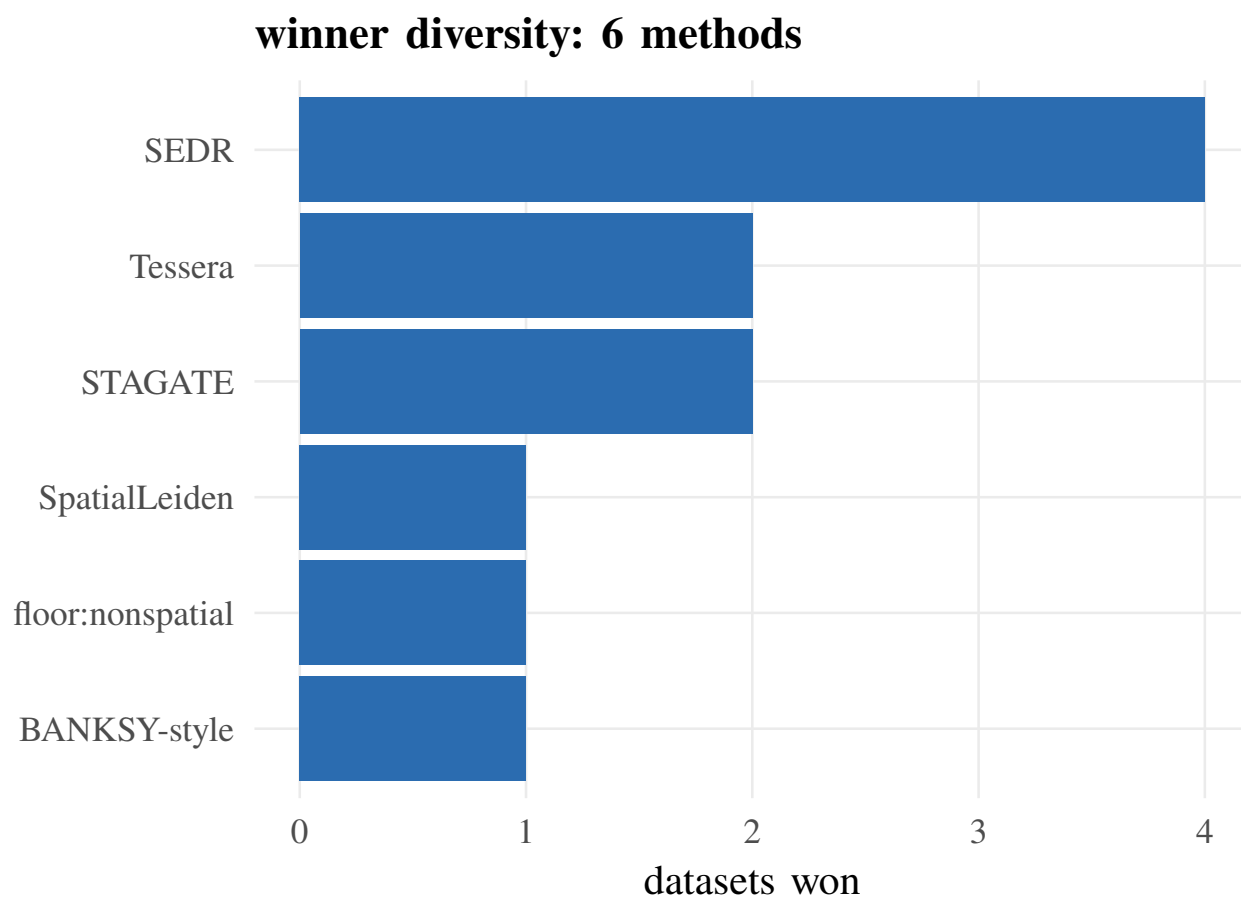


Fig. 9: The law survives the author-recommended backend (real R mclust): published versus mclust-backend advantage stays close to the diagonal for most datasets (a; a few, such as IMC, move more), and the fairer backend shifts every method's *mean* ARI by ≤ 0.016 (b).

VIII. BACKEND FAIRNESS: THE LAW IS NOT A BACKEND ARTIFACT

Re-running all 11 datasets with the clusterer the deep methods actually recommend (R's `mclust`) leaves the conclusions intact and slightly strengthens the law, to $\rho = +0.79$ ($p = 0.004$, Fig. 9), above the published GMM-tied $+0.72$. Real `mclust` wins 20 of 81 method $\times$ dataset cells but shifts each method's mean ARI by at most 0.016. The recommended backend makes the law cleaner, not weaker.

IX. A WORKED EXAMPLE: A FALSE SOTA AND A FALSE MECHANISM FROM ONE DATASET

We close with the candidate boundary-aware method (Tessera) as one worked example of how single-dataset evaluation can certify two claims that do not survive scrutiny: a state-of-the-art ranking, and a working novel mechanism. We do not claim every SOTA report is false; we show, on a method we built and can fully dissect, how both claims arise and how a multi-dataset, ablation-controlled view overturns them.

a) A single favourable dataset certifies a false SOTA.:

On `osmFISH` the method ranks first (ARI 0.438, $+0.196$ over

the non-spatial floor), a publishable state-of-the-art claim from a single ARI table. Across the full 11-dataset panel, however, it is a backend-robust generalist, not a winner: its rank swings from 1 to 8 (Fig. 2), its one win (`osmFISH`) is by only 0.005 ARI over the next spatial method, and on IMC it trails the best method by 0.39 ARI. On the other contiguous task, `MERFISH`, the top spot is a seed-level tie it does not hold (the winner flips `SpaceFlow`, `Tessera`, `SpaceFlow` across three runs). No dataset has it uniquely best; its real, and far more modest, property is backend stability: $\Delta\text{ARI}(\text{KMeans} \leftrightarrow \text{GMM}) = 0.036$, descriptively rank 1/7 in the DLPCF comparison, where the other methods' shifts are approximately 0.10–0.20.

b) The headline mechanism does not drive the gain.:

The method's advertised contribution is anti-over-smoothing through the learned edge gate (Fig. 10). A component ablation on the two datasets where it is competitive (Fig. 11) refutes that account: only the boundary-contrastive loss matters. Removing it collapses ARI $0.45 \rightarrow 0.17/0.20$, back to a plain graph-autoencoder backbone, while the named edge gate and multi-scale fusion are neutral-to-harmful (-0.01 to -0.03). The full model beats the backbone by $+0.29/+0.23$, but that entire gain is the generic contrastive objective, not the mechanism the method is named for.

c) *The verdict.:* Single-dataset evaluation would have certified two false positives here: a state-of-the-art ranking and a working new mechanism. The full panel and the ablation overturn both. This is the failure mode the predictive law pre-empts: decide whether a spatial prior helps at all before crediting any one method, or any one of its components, for the gain.

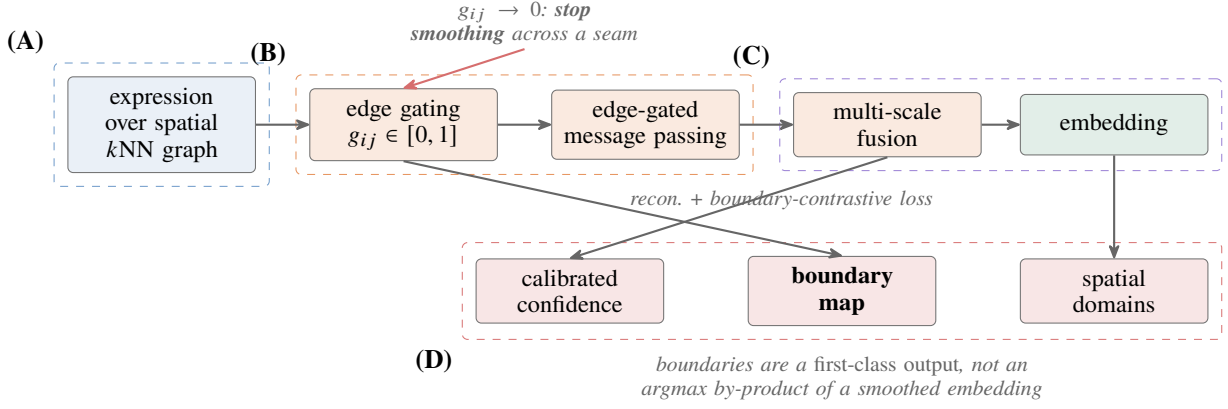


Fig. 10: The candidate method (Tesseract) in one diagram. A learned per-edge gate g_{ij} down-weights message passing across tissue seams (anti-over-smoothing); multi-scale fusion combines the gated representations into an embedding that is clustered into domains, with the boundary map and a calibrated confidence as explicit outputs. The component ablation (Fig. 11) tests whether each part contributes.

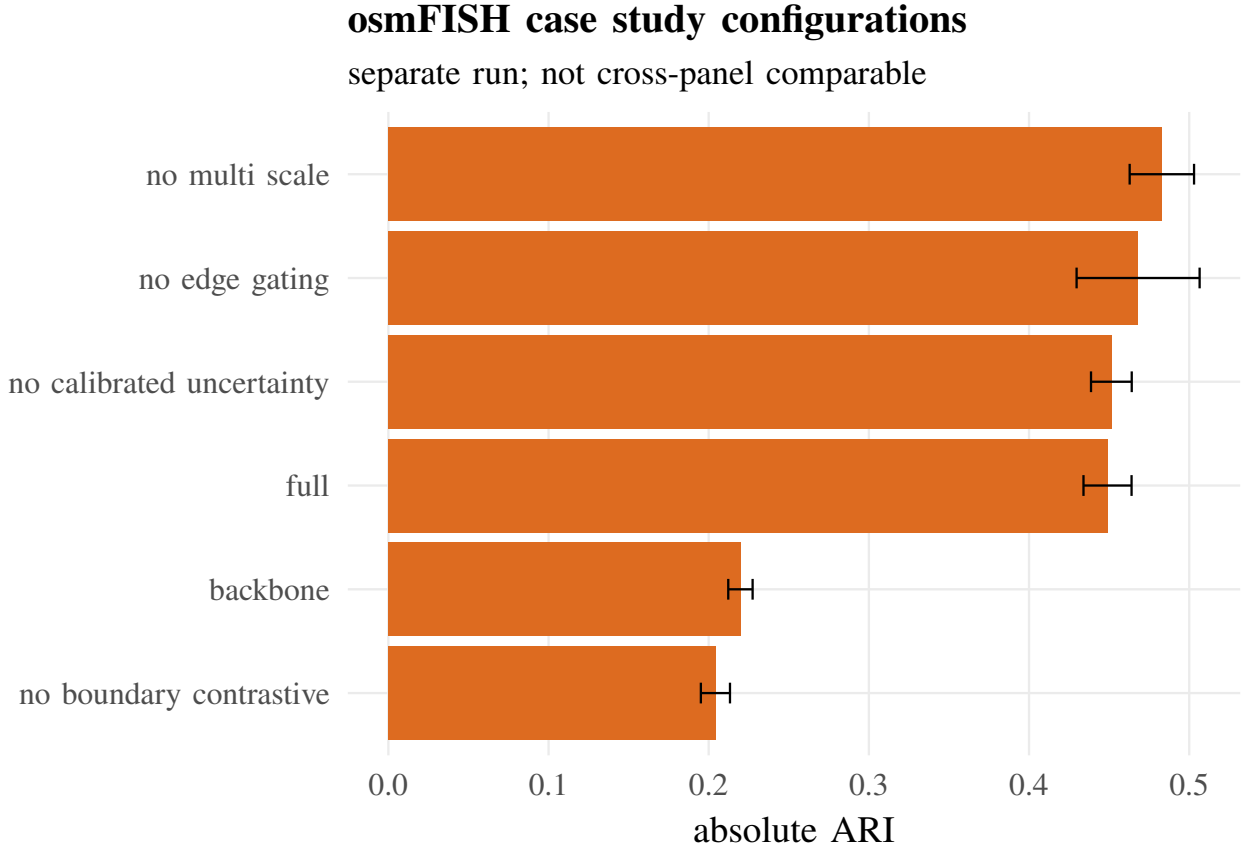


Fig. 11: Component ablation of the candidate method on the two datasets where it is competitive (3 seeds): the ARI drop when each component is removed (positive = the component contributes). Only the boundary-contrastive loss matters; the headline anti-over-smoothing edge gate and multi-scale fusion are neutral-to-harmful, so the model’s gain over a plain backbone is the contrastive loss alone. These absolute ARIs come from a separate two-dataset case-study run and are not directly comparable with the cross-dataset benchmark panels.

X. DISCUSSION

The unit of progress in spatial-domain detection should shift from “a better method” to “a better understanding of when spatial priors help”. Our law turns the discouraging “no universal SOTA” finding into an actionable, label-free check made before committing: estimate the spatial-coherence gain

on the unlabelled data; if it is low, a non-spatial clusterer is both floor and ceiling and spatial smoothing can actively hurt (this direction is demonstrated causally in the synthetic manipulation of §VII, where the advantage falls to -0.33 to -0.34 ; on the real low-contiguity datasets in our panel the effect is present but at seed-level magnitude, spatial advantage

−0.005 on seqFISH and Open-ST and +0.008 to +0.024 on Slide-seqV2, MIBI-TOF, and CODEX, Table IV); if it is high, a spatial method is warranted, though which one is within seed noise. The inverted profile of SpatialLeiden, and the fact that a popular deep method (SpaGCN) wins only a single scattered dataset, both reinforce that method–task fit, not method identity, is what determines performance.

Because the relationship is causal, not merely correlational, and does not stand in for cluster count, dimensionality, or modality, it can be used to choose a method in advance rather than to rationalise a result after the fact. Removing the labels costs significance but keeps a positive predictor; broadening to noisier tasks shrinks the correlation. Neither overturns the rule.

The benchmark also identifies backend robustness as a distinct, secondary axis of method quality. Tesseract is not the SOTA label-agreement method, but its $\Delta\text{ARI}(\text{KMeans} \leftrightarrow \text{GMM})$ of 0.036 is descriptively rank 1/7 for backend stability, versus shifts of approximately 0.10–0.20 for the more sensitive methods. This does not establish statistical superiority; it shows why robustness to an often-unreported analysis choice should be reported alongside accuracy rather than hidden by a single backend.

XI. LIMITATIONS

a) Sample size and panel diversity.: The headline rests on $n=11$ technologically diverse, de-duplicated datasets; broadening to three additional cell-type cancer datasets ($n=14$) attenuates the correlation from +0.72 to +0.56 ($p = 0.038$), so +0.72 is a diverse-panel upper estimate, not a constant, and high-contiguity domain-GT datasets remain scarce.

b) Does continuous contiguity beat a coarse GT-type label?: A cheaper alternative to measuring contiguity would be a binary label read straight off Table II’s GT-type column: structural ground truth (domain/region/layer/tumour) versus identity-based ground truth (cell type/niche/annotation/ cluster). On the same 11-dataset panel this coarse indicator correlates with spatial-prior advantage at $\rho=0.58$ ($p = 0.062$, point-biserial $r = 0.52$, $p = 0.099$) – weaker and no longer significant at $\alpha=0.05$, versus $\rho=0.72$ ($p = 0.012$) for continuous contiguity. The gap is driven by Slide-seqV2 (region GT, measured contiguity only 0.274) and IMC (cell-type GT, measured contiguity 0.630): both are exceptions the categorical split misclassifies, because GT-type name is only a weak proxy for how spatially contiguous the labels actually are on a given tissue. Continuous, data-measured contiguity is therefore not redundant with a coarser categorical predictor built from the same metadata.

c) Multiple comparisons and influence.: The headline $p = 0.012$ is uncorrected and the estimate is fragile to its small sample: it survives a Bonferroni correction over the three primary tests ($p = 0.035$) but not over the wider family of seven predictor–advantage correlations ($p = 0.082$), and dropping the single most influential dataset (MERFISH) lowers ρ to 0.63 (jackknife range [0.63, 0.75]). Only the advantage correlation is significant; the method-specific rank correlations are not. The out-of-sample decision rule is real but marginal at $n=11$ ($p =$

0.06), and the label-free `coh_gain` proxy is directional only: after correcting for its selection among six candidate proxies it is not significant (search-corrected $p = 0.28$; asymptotic uncorrected $p = 0.079$), so the causal synthetic sweep, not the proxy, carries the weight of the label-free claim. The primary family of three tests comprises the correlations against the study’s actual predictive targets reported in Table V (contiguity vs. spatial-prior advantage, vs. Tesseract rank, vs. SpaceFlow rank); the wider family of seven additionally includes confound-marginal correlations (cluster count, dimensionality, modality) and two label-free proxy validations (Moran’s I , raw coherence) computed for different diagnostic purposes, not repeated tests of the same hypothesis.

d) Partly definitional direction.: The advantage we predict and the contiguity predictor are both read off the same spatial k NN structure on the same labels. On this panel the neighbour-mean smoother is never the winning spatial method, so the advantage never depends on it specifically; but every spatial method here, learned graph encoders and a neighbour-augmentation method alike, is built on that same k NN structure, so “spatial methods help when the labels are spatially contiguous” is closer to a calibration than a discovery. What is genuinely empirical is *how much* contiguity is needed before a prior pays off, together with the out-of-sample and label-free results above. The conclusion has already changed twice as the panel grew; it is the current best estimate, not a final word.

e) Baseline tuning and metrics.: SpaGCN is run with a kmeans initialisation (its native louvain path is blocked by the environment) and so its absolute level is understated; GraphST receives the recommended R mclust backend on the datasets where it runs (size-gated to $\leq 8k$ cells) but not its full native preprocessing/refinement tuning, so its absolute level is a lower bound. Because GraphST anchors the low end of Table III, the headline “backend-shift equals between-method spread” framing of §IV-A depends on retaining this under-tuned baseline: dropping GraphST shrinks the spread from 0.203 to 0.150, below the largest single-backend shift. SpaceFlow additionally receives its own native normalise/log/HVG/PCA pipeline rather than the shared scaled expression matrix used by the graph methods – a defensible per-method choice but a variable not held fixed across methods, and SpaceFlow drives part of the headline correlation (MERFISH winner, Table IV). All metrics are semi-circular or conditioned: no metric here is external truth.

DATA AVAILABILITY

All datasets are previously published and publicly available from their original sources: DLPFC [8], seqFISH [25], MERFISH [2], STARmap [26], osmFISH [27], MIBI-TOF [28], breast IMC [29], a breast-tumour Visium section [30] with SEDR pathology-domain labels [4], Open-ST [31], Slide-seqV2 [32], and CODEX [33]; several are redistributed through Squidpy [24]. No new data were generated. Accession identifiers and download locations for each dataset (repository DOIs, GEO/Single-Cell-Portal accessions, or the originating lab’s data portal, as applicable) are listed in the code repository, alongside the original publications cited above.

CODE AVAILABILITY

All analysis code, the synthetic-tissue generator, and the verification harness are available at a public repository (<https://github.com/PeterPonyu/tessera-st>) and archived at Zenodo (DOI assigned on deposit). Every reported number is regenerated from the analysis scripts and machine-checked against its JSON artifact by an automated verification harness; a companion robustness-audit routine recomputes the multiple-comparison corrections, the jackknife, the out-of-sample binomial test, and the smoother-independence check from the same artifacts. Figures are vector graphics generated by an R script that reads the JSON artifacts directly, builds the synthetic spatial-domain map, and emits the `pgfplots` panels, the heatmap, and the result tables (compiled with `pdflatex`); the whole pipeline is reproducible with a single build command. Baseline method brand names are confined to driver scripts (a strict separation enforced by automated tests). The repository also ships an audit log documenting the full evaluation history, including one data-provenance issue caught by internal code review and fixed by a full genuine re-run.

AUTHOR CONTRIBUTIONS

Z.F. conceived the study, implemented the methods and benchmark, performed the analyses, and wrote the manuscript.

COMPETING INTERESTS

The author declares no competing financial or personal interests.

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